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MTH621 Real Analysis I

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If f is differentiable at x_0 , then prove that f is continuous at x_0 .

Answer ([Please click here to Add Answer](#))

4.1.1 Differentiability Implies Continuity

Theorem: If f is differentiable at x_0 , then f is continuous at x_0 .

Proof:

$$f(x) = f(x_0) + [f'(x_0) + E(x)](x - x_0),$$

where E is defined on a neighborhood of x_0 and

$$\lim_{x \rightarrow x_0} E(x) = E(x_0) = 0.$$

where

$$E(x) = \begin{cases} \frac{f(x) - f(x_0)}{x - x_0} - f'(x_0), & x \in D_f \text{ and } x \neq x_0, \\ 0, & x = x_0. \end{cases}$$

Examples:

State the chain rule for differentiating compositions of functions

Theorem (The chain rule): Suppose that g is differentiable at x_0 and f is differentiable at $g(x_0)$.

Then the composite function $h = f \circ g$, defined by

$$h(x) = f(g(x)),$$

is differentiable at x_0 , with

$$h'(x_0) = f'(g(x_0))g'(x_0).$$

Proof: Since f is differentiable at $g(x_0)$, we can write

$$f(t) - f(g(x_0)) = [f'(g(x_0)) + E(t)][t - g(x_0)],$$

where

$$\lim_{t \rightarrow g(x_0)} E(t) = E(g(x_0)) = 0. \quad (4.3)$$

4.1. Derivative

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Letting $t = g(x)$ yields

$$f(g(x)) - f(g(x_0)) = [f'(g(x_0)) + E(g(x))][g(x) - g(x_0)].$$

Since $h(x) = f(g(x))$, this implies that

$$\frac{h(x) - h(x_0)}{x - x_0} = [f'(g(x_0)) + E(g(x))]\frac{g(x) - g(x_0)}{x - x_0}. \quad (4.4)$$

Since g is continuous at x_0 we have

$$\lim_{x \rightarrow x_0} E(g(x)) = E(g(x_0)) = 0.$$

Therefore, (4.4) implies that

$$h'(x_0) = \lim_{x \rightarrow x_0} \frac{h(x) - h(x_0)}{x - x_0} = f'(g(x_0))g'(x_0),$$

as stated.

State Rolle's Theorem .

4.5 Rolle's Theorem

Theorem: Suppose that f is continuous on the closed interval $[a, b]$ and differentiable on the open interval (a, b) , and $f(a) = f(b)$. Then $f'(c) = 0$ for some c in the open interval (a, b) .

Proof: Since f is continuous on $[a, b]$, f attains a maximum and a minimum value on $[a, b]$. If these two extreme values are the same, then f is constant on (a, b) , so $f'(x) = 0$ for all x in (a, b) .

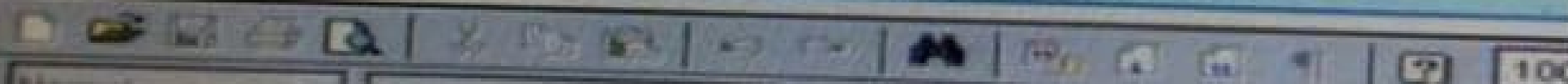
If the extreme values differ, then at least one must be attained at some point c in the open interval (a, b) , and $f'(c) = 0$.

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Determine whether the function $f(x) = 1 + x$ is integrable on $[a, b]$?

Answer ([Please click here to Add Answer](#))



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Example: Determine whether the function $f(x) = 1 + x$ is integrable on $[a, b]$?

Solution: If

$$f(x) = \sin x \quad \text{and} \quad g(x) = \frac{1}{x}, \quad x \neq 0,$$

then

$$h(x) = f(g(x)) = \sin \frac{1}{x}, \quad x \neq 0,$$

and

$$h'(x) = f'(g(x))g'(x) = \left(\cos \frac{1}{x} \right) \left(-\frac{1}{x^2} \right), \quad x \neq 0.$$